

The Bill Belongs to the Fan, Not the Path

Anti-Diagonal Causalization and Counterfactual Cost Conservation

Abstract

We isolate a simple causal principle in exact zero repair. Every static zero-repair decomposition admits a lossless anti-diagonal causal section in the coordinate $s = j + y$. Projecting this section to first moments and subtracting the optimal value for the remaining horizon yields an exact conservation law for a nonnegative *frontier defect*. At a repeated frontier contact the realised continuation carries zero excess, while the entire marginal cost of the longer horizon is carried by the unrealised siblings. Thus the causal bill is a property of the fan of simultaneously maintained futures, not of the realised path alone. We finally compare this identity with a distinct horizon anti-diagonal construction in which genuine dependence is present globally but cannot be jointly exposed by any admissible path. The two mechanisms are not identified numerically; they share the same causal separation between global structure and pathwise visibility.

1 Exact anti-diagonal causalization

Let P, Q be probability laws of finite mean on $\mathcal{A} \subseteq \mathbb{Z}_{\geq 0}$ with

$$\mathbb{E}P = \mathbb{E}Q. \quad (1)$$

The motivating quartic pair is

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right),$$

for which $G_P(z) - G_Q(z) = (1 - z)^4/64$, but the conservation theorem below only uses (1).

At a zero-repair node let B_h be the current reserve law and B_{hx} the child law after output x . Exactness means

$$Q * B_h = \sum_{x \in \mathcal{A}} P(x) \delta_x * B_{hx}. \quad (2)$$

Lemma 1 (Lossless anti-diagonal section). *Put*

$$S_h(s) := (Q * B_h)(s).$$

For $S_h(s) > 0$ define

$$K_h(x \mid s) := \frac{P(x)B_{hx}(s - x)}{S_h(s)}, \quad (3)$$

with $B_{hx}(k) = 0$ for $k < 0$. Then $K_h(\cdot \mid s)$ is a probability law and

$$z_h(j, y, x) := B_h(j)Q(y)K_h(x \mid j + y) \quad (4)$$

is an exact causal coupling. In particular,

$$P(x)B_{hx}(k) = S_h(k + x)K_h(x \mid k + x). \quad (5)$$

Proof. Taking the coefficient of s in (2) gives

$$S_h(s) = \sum_x P(x)B_{hx}(s - x),$$

so (3) sums to one. Equation (4) preserves the fresh $B_h \otimes Q$ marginal. Moreover,

$$\Pr(X = x, J' = k) = \sum_{j+y=k+x} B_h(j)Q(y)K_h(x \mid k + x) = S_h(k + x)K_h(x \mid k + x) = P(x)B_{hx}(k),$$

which proves both the P marginal and the prescribed child laws. $\square$

Thus $s = j + y$ is not a heuristic compression: it is an exact quotient of the local causal geometry.

2 The fan conservation theorem

Let $\mathcal{K}_r$ be the set of roots of exact depth- r trees and define

$$\mathcal{B}_r := \inf_{B \in \mathcal{K}_r} \mathbb{E}B, \quad \mathcal{B}_0 := 0.$$

For a node h in a depth- n tree write $r(h) = n - |h|$. Since the subtree below h is feasible for the remaining horizon, define the nonnegative *frontier defect*

$$\Delta_h := \mathbb{E}B_h - \mathcal{B}_{r(h)} \geq 0. \quad (6)$$

For a word $g = (g_1, \dots, g_s)$ put $P(g) = \prod_i P(g_i)$.

Theorem 2 (Frontier-defect conservation). *If h has r steps remaining, then for every $0 \leq s \leq r$,*

$$\boxed{\sum_{|g|=s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.} \quad (7)$$

Equivalently, the exponentially many descendant defects form a weighted decomposition of one scalar excess above the shorter-horizon frontier.

Proof. Taking first moments in (2) and using (1),

$$\mathbb{E}B_h = \sum_x P(x) \mathbb{E}B_{hx}. \quad (8)$$

Iterating (8) for s generations gives

$$\mathbb{E}B_h = \sum_{|g|=s} P(g) \mathbb{E}B_{hg}.$$

Every descendant hg has remaining horizon $r - s$, hence by (6)

$$\mathbb{E}B_{hg} = \mathcal{B}_{r-s} + \Delta_{hg}.$$

Therefore

$$\mathbb{E}B_h = \mathcal{B}_{r-s} + \sum_{|g|=s} P(g) \Delta_{hg}.$$

But also $\mathbb{E}B_h = \mathcal{B}_r + \Delta_h$. Equating the two expressions proves (7). $\square$

The same identity can be read directly as a cost projection of the anti-diagonal section. Multiplying (5) by $k = s - x$ and summing over s gives

$$P(x) \mathbb{E}B_{hx} = \sum_s (s - x) S_h(s) K_h(x | s). \quad (9)$$

Subtracting $P(x) \mathcal{B}_{r-1}$ and summing over x yields the one-step form of (7):

$$\boxed{\sum_x P(x) \Delta_{hx} = \Delta_h + d_r, \quad d_r := \mathcal{B}_r - \mathcal{B}_{r-1}.} \quad (10)$$

Hence frontier-defect conservation is precisely the first-moment cost projection of the exact anti-diagonal causal disintegration.

Corollary 3 (The bill belongs to the fan). *Suppose h is on the optimal frontier, $\Delta_h = 0$, and one child hx_* is again on the optimal frontier, $\Delta_{hx_*} = 0$. Then*

$$\boxed{d_r = \sum_{x \neq x_*} P(x) \Delta_{hx}.} \quad (11)$$

Thus the realised contact branch carries no excess at all; the complete marginal price of preserving the longer horizon is carried by the alternative branches that had to remain feasible simultaneously.

Proof. Set $s = 1$ in (7) and remove the two zero terms. $\square$

There is no paradox here. A path is one realised continuation; causal feasibility is a constraint on the whole fan. Equation (11) says exactly where the difference lives. One may therefore summarize the accounting rule without introducing a nomenclature long enough to require a separate index of academic titles:¹

The bill belongs to the fan, not the path.

¹A deliberately small tribute to the satirical Henryk Lahola tradition: here one scalar identity is sufficient.

3 A second anti-diagonal: hiding beyond the horizon

The previous anti-diagonal is the local coordinate $s = j + y$. A distinct anti-diagonal mechanism appears in causal information trees. Its role is not to account for reserve, but to hide globally present dependence from every admissible path.

Let $n = mL$. Split streams $X_{1:n}$ and $W_{1:n}$ into blocks of length L and correlate $W^{(j)}$ with $X^{(j+1)}$, $j = 1, \dots, m - 1$. Read the X stream forward and the W stream in reverse.

Proposition 4 (Horizon anti-diagonal hiding). *No action history of length at most n can expose both a coordinate of $X^{(j+1)}$ and its correlated partner in $W^{(j)}$.*

Proof. To expose a coordinate of $X^{(j+1)}$ requires at least $jL + 1$ forward X -requests. Through the reversed W stream, exposing its partner in $W^{(j)}$ requires at least $n - jL + 1$ W -requests. Seeing both therefore requires at least

$$(jL + 1) + (n - jL + 1) = n + 2$$

actions, which exceeds the horizon. □

The two anti-diagonals should not be conflated. Proposition 4 does not identify hidden mutual information with the reserve defect Δ_h . What is common is the causal architecture:

global structure can be nonzero
while the realised path does not carry or expose it.

In the information tree, the anti-diagonal places the two ends of a dependence outside every admissible path. In zero repair, the anti-diagonal section splits exact mass among branch channels, and Theorem 2 gives the exact accounting law for their excess cost.

Conclusion

The connection is therefore algebraic on the zero-repair side and structural across the two causal models:

anti-diagonal disintegration $\xrightarrow{\text{first moment--optimal frontier}}$ counterfactual cost conservation.

At frontier contacts this becomes the sharp statement (11): the realised continuation pays zero excess, while the fan pays exactly the marginal bill. The remaining open question is not whether this mechanism exists, but how its forced redistribution controls the asymptotic growth of $\mathcal{B}_n$.